

CO3-AT3- ERROR ANALYSIS

Quantizer → Decimator → Interpolator → Polyphase Filter

can produce aliasing, oscillations, amplitude distortion, and unwanted spectral components if the stages are implemented incorrectly. The errors and corrections can be analyzed stage by stage.

1. Quantizer

Possible implementation errors

- Quantization is performed without selecting an appropriate number of bits.
- Input signal exceeds the quantizer range, causing clipping.
- Quantization noise is ignored.
- The quantizer output is not properly scaled before further processing.

Correction

- Select sufficient quantization bits, e.g., 8–16 bits depending on the required SNR.
- Normalize the input so that it remains within the quantizer range.
- Account for quantization noise in the system design.

Justification:

Clipping introduces nonlinear distortion, while insufficient quantization resolution increases quantization noise. Proper scaling preserves signal quality before decimation.

2. Decimator

Suppose the signal is downsampled by a factor **M**.

Major error: Directly downsampling without an anti-aliasing filter.

For decimation,

$$\begin{bmatrix} x_d[n] = x[nM] \end{bmatrix}$$

Before downsampling, the signal must be low-pass filtered with cutoff approximately

$$\begin{bmatrix} f_c \leq \frac{f_s}{2M} \end{bmatrix}$$

where (f_s) is the original sampling frequency.

Correction

Use:

$$[x[n] \rightarrow \text{LPF} \rightarrow \downarrow M]$$

The anti-aliasing filter must strongly attenuate frequencies above the new Nyquist frequency.

Justification:

Downsampling compresses the frequency spectrum. Components above the new Nyquist frequency fold into the baseband, producing irreversible aliasing.

3. Interpolator

For interpolation by a factor **L**, the signal is first upsampled:

$$[x_i[n] = \begin{cases} x[n/L], & n=0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}]$$

Implementation errors

- Zero insertion is performed without an interpolation filter.
- The filter cutoff is incorrect.
- Filter gain is not compensated after interpolation.
- The filter is not designed for the new sampling frequency.

Correction

Use:

$$[x[n] \rightarrow \uparrow L \rightarrow \text{LPF}]$$

The interpolation filter should have a cutoff near

$$[f_c \leq \frac{f_s}{2L}]$$

with appropriate gain compensation, commonly a gain of approximately (L).

Justification:

Zero insertion creates spectral images around multiples of the original sampling frequency. The interpolation low-pass filter removes these images and reconstructs a smooth waveform.

4. Polyphase Filter

Possible implementation errors

- Incorrect decomposition of the filter coefficients into polyphase components.
- Wrong phase ordering.
- Incorrect delay alignment.
- Polyphase branches use unequal or incorrect coefficients.
- Filter is applied at the wrong sampling rate.
- Transition-band specifications are inadequate.

For an interpolation factor (L), a filter can be decomposed as

$$[\\ H(z)=\\sum_{k=0}^{L-1} z^{-k} E_k(z^L) \\]$$

where ($E_k(z)$) are the polyphase components.

Correction

- Divide the FIR filter coefficients correctly among the (L) phases.
- Maintain the correct phase order.
- Verify delays and coefficient indexing.
- Design the prototype filter using the required passband, stopband and transition width.
- Ensure sufficient stopband attenuation.

Justification:

A wrongly implemented polyphase structure can produce phase errors, amplitude ripple and oscillations even when the original FIR filter is correct.

5. Source of Oscillations

The observed oscillations can mainly result from:

1. **Insufficient stopband attenuation**
2. **Incorrect filter cutoff frequency**
3. **Too-narrow/poorly designed transition band**
4. **Incorrect polyphase coefficient ordering**
5. **Phase or delay mismatch between branches**
6. **Sharp truncation of FIR coefficients**

These can produce **passband ripple and ringing**, particularly near sharp signal transitions.

A better FIR filter can be designed using a window, Kaiser method, Parks–McClellan algorithm, or an appropriately designed equiripple filter.

6. Corrected Complete Processing Chain

The proper multirate architecture should be:

```
[
\boxed{
\text{Input}
\rightarrow
\text{Quantizer}
\rightarrow
\text{Anti-Alias LPF}
\rightarrow
\downarrow M
\rightarrow
\uparrow L
\rightarrow
\text{Interpolation/Polyphase LPF}
\rightarrow
\text{Output}
}
]
```

For rational sampling-rate conversion,

```
[
f_{out}=f_{in}\frac{L}{M}
]
```

the preferred practical structure is:

```
[
\boxed{
x[n]\rightarrow\uparrow L\rightarrow
\text{Polyphase LPF}\rightarrow\downarrow M
}
]
```

This avoids explicitly generating unnecessary zero-valued samples.

Error–Correction Summary

Stage	Implementation error	Correction	Effect
Quantizer	Too few bits/clipping	Increase resolution and normalize	Reduces quantization distortion
Decimator	No anti-aliasing LPF	LPF before $\downarrow M$	Prevents aliasing
Decimator	Wrong cutoff	$(f_c \leq f_s/(2M))$	Preserves valid spectrum
Interpolator	No reconstruction LPF	LPF after $\uparrow L$	Removes spectral images
Interpolator	Wrong filter gain	Apply proper gain compensation	Maintains amplitude

Stage	Implementation error	Correction	Effect
Polyphase	Wrong phase ordering	Correct coefficient decomposition	Removes phase distortion
Polyphase	Insufficient attenuation	Increase filter order/stopband attenuation	Reduces unwanted components
Filter	Excessive passband ripple	Proper FIR design	Reduces oscillations
Overall	Incorrect sampling-rate calculations	Verify $(f_{\text{out}} = f_{\text{in}} L/M)$	Ensures correct conversion

Conclusion

The primary cause of aliasing is inadequate anti-aliasing filtering before decimation, while the primary causes of spectral images and oscillations are incorrect interpolation filtering, poor FIR design, and polyphase phase/coefficient errors. The system should use properly designed low-pass filters, correct sampling-rate relationships, sufficient quantization resolution, and correctly ordered polyphase branches. This produces a stable output with minimal aliasing, reduced oscillations, and accurate amplitude and frequency characteristics.